RAJIV GANDHI PROUDYOGIKI VISHWAVIDYALAYA, BHOPAL

DIPLOMA IN MINE SURVEYING

SEMESTER: FOURTH SCHEME: July 2012
COURSE CODE: 401 PAPER CODE: 6615
NAME OF COURSE: **MINE SURVEYING DRAWING**

COMMON WITH PROGRAM (S):

RATIONALE

The important job function of Mine surveyor is to prepare a mine plan & section with required precisions and accuracy is given in mine regulation. Perfect, accurate and good plan is always update with upgrades as time demand to avoid the discrepancy of works. Mine office should have to keep every surveying instrument & apparatus. Drawing of any symbol for mine working helps to easily understand the feature of mine.

On this course student will be able to:

1. Know the scale of mine plans & section
2. Layout the mine office, & prepare different type of mine plan.
3. Develop the drawing skill of mine sign & symbol.
4. Knowledge of storages & maintenances of mine plans.
5. Solve the plotting problems of co-ordinates with scale of plan.
6. Solve the problems with graphically related to dip, borehole, drift and faults.

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DIPLOMA IN MINE SURVEYING

SEMESTER: FOURTH SCHEME: July 2013
COURSE CODE: 401 PAPER CODE: 6615
NAME OF COURSE: MINE SURVEYING DRAWING
COMMON WITH PROGRAM (S):

SCHEME OF STUDIES

S. No.	TOPICS	THEORY (HOURS)
1.	Scales, Survey Office, & Mine Models	15
2.	Signs & Symbols;	15
3.	Plotting Of Survey Work;	15
4.	Plotting Problem;	15
5.	Plans & sections	15
6.	Enlargement & Reduction of Plans & sections :	15
	TOTAL	90

RAJIV GANDHI PROUDYOGIKI VISHWAVIDYALAYA, BHOPAL

DIPLOMA IN MINE SURVEYING

SEMESTER: FOURTH SCHEME:
July 2012
COURSE CODE: 401 PAPER CODE:
6615
NAME OF COURSE: **MINE SURVEYING DRAWING**

COMMON WITH PROGRAM (S):

Lectures: **6** Hrs. per week

CONTENT DETAILS

S. No.	Course Contents	Hrs of Study
01	**Scales, Survey Office, & Mine Models**; • Scales & its Classification • Survey office layout • Concept of Mine Models • Type of models	15
02	**Signs & Symbols;** • Drawing Sign & Symbols as given in the II Schedule of CMR-1957. • Drawing Sign & Symbols as given in the II Schedule of MMR-1961.	15
03	**Plotting Of Survey Work;** • Plotting by Scales & Protractor method • Plotting by Coordinate System • Advantages & disadvantages of both methods and their comparisons.	15
04	**Plotting Problem;** • Coordinate system • Dip, drift, fault • Borehole	15
05	**Plans & sections;** • List of plans & sections	15

	• Preparation of each plans & section • Care & maintenance of mines plans. • Storage of plans & sections • Preservation of plans • Keeping of records • Checking accuracy of Old Plans.	
06	**Enlargement & Reduction of Plans & sections;** • Method of Enlargement & Reduction of Plans & sections • Different type of instrument uses for Enlargement & Reduction of Plans & sections	15

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DIPLOMA IN MINE SURVEYING

SEMESTER: FOURTH SCHEME:
July 2012
COURSE CODE: 401 PAPER CODE:
6615
NAME OF COURSE: **MINE SURVEYING DRAWING**

COMMON WITH PROGRAM (S):

REFERENCE BOOKS

1. **Coal Mine Regulation 1957** **L.C.Kaku**
2. **Metal Mine Regulation 1961** **L.C.Kaku**
3. **Mine Surveying Vol. - I, II & III** **S. Ghatak**
4. Surveying Vol. - I, II

RAJEEV GANDHI PROUDYOGIKI VISHWAVIDYALAYA, BHOPAL

DIPLOMA IN MINE SURVEYING

SEMESTER: FOURTH SCHEME: July2012

COURSE CODE: 402 PAPER CODE: 6616

NAME OF COURSE: **ELEMENTS OF CIVIL ENGINEERING**

COMMON WITH PROGRAM (S):

RATIONALE

The Civil works is most common to each and every field of engineering. Civil is a basic engineering subject and is a prerequisite to any other project. Before starting any project of engineering such as electrical, mechanical, mining or metallurgy, civil works are required like planning township, plant construction, water supply and other similar works. In Mining or Mine surveying Engineering, knowledge of civil engineering is required in surface planning, roads, building, underground construction of stopping, Dams, etc.

After undergoing the course of study the student shall be able to

1. Know the material technology that is used in general civil work.
2. Provide appropriate surveying alignment for road and rail setup.
3. Affiliate them with work of general construction detail of Dam
4. Enhance the skill of quantity measurement.

RAJIV GANDHI PROUDYOGIKI VISHWAVIDYALAYA, BHOPAL

DIPLOMA IN MINE SURVEYING

SEMESTER: FOURTH SCHEME: July2012

COURSE CODE: 402 PAPER CODE: 6616

NAME OF COURSE: **ELEMENTS OF CIVIL ENGINEERING**

COMMON WITH PROGRAM (S):

SCHEME OF STUDIES

S. No.	TOPICS	THEORY (HRS.)
7.	GENERAL MATERIAL TECHNOLOGY	20
8.	CONSTRUCTION TECHNOLGY	20
9.	ROAD ALIGNMENT & RAIL ALIGNMENT	15
10.	SIMPLE CONSTRUCTION DETAIL OF BRIDGE & DAM	20
11.	QUANTITY MEASUREMENT	15
	TOTAL	90

RAJIV GANDHI PROUDYOGIKI VISHWAVIDYALAYA, BHOPAL

DIPLOMA IN MINE SURVEYING

SEMESTER: FOURTH SCHEME:
July2012
COURSE CODE: 402 PAPER CODE: 6616
NAME OF COURSE: **ELEMENTS OF CIVIL ENGINEERING**

COMMON WITH PROGRAM (S):

Lectures: **6** Hrs. per week

CONTENT DETAILS

S. No.	Course Contents	Hrs of Study
01	GENERAL MATERIAL TECHNOLGY; • BRICKS: Type of bricks, good quality of bricks, Laboratory tests for bricks. • STONE AGGREGATES; Requirement of good quality of stones, test for stones, selection of stone for different civil works, coarse & fine agreements, Fitness modules • CEMENT; Composition of cement, Type of cement, Laboratory test & Field test on cement. • STEEL & STEEL PRODUCT; Composition of steel, change in properties due to alloying ,impurities in steel, steel alloy, defect in steel, steel section, test of M.S.Bar. • TIMBER & TIMBER PRODUCT; Characteristics of good timber, defect in timber, preservatives, seasoning of timber.	20
02	CONSTRUTION TECHNOLOGY: • FOUNDATION; Necessity of foundation, bearing capacity & safe bearing capacity, type of foundation, shallow foundation, deep foundation, selection of type of foundation, procedure to give layout, different terms, centre line plan, foundation plan, checking the accuracy of layout, utility of control point. • MASONARY; Type of masonry, necessity bond of bricks masonry, type of stone masonry, dry stone masonry, revetment. • CONCRETE; Ingredients in cement concrete properties of	20

		ingredient, grade & strength of concrete, nominal mix controlled concretes, work ability of concretes, slum test, water cement ratio, test for compressive strength of concrete, factors which affect the quality of concrete, use of reinforcement in concrete, R.C.C.& P.C.C.	

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DIPLOMA IN MINE SURVEYING

SEMESTER: FOURTH SCHEME: July2012

COURSE CODE: 402 PAPER CODE: 6616

NAME OF COURSE: **ELEMENTS OF CIVIL ENGINEERING**

COMMON WITH PROGRAM (S):

Lectures: **6** Hrs. per week

CONTENT DETAILS

03	ROAD ALIGNMENT; • Type of Roads • General detail of Roads Construction; • Topographical survey; Establishing Survey Control Points, Levelling, Detailed Topographical survey, Cross-Section Surveys. • Design Criteria; Geometric Design, Pavement Design, Drainage. RAIL ALIGNMENT; • Survey for track alignment. • Traffic Survey. • Reconnaissance survey-Instrument. • Preliminary survey- Instrument. • Detail survey or final location survey	15
04	SIMPLE CONSTRUCTION DETAIL OF BRIDGES & DAMS; BRIDGES; • Components of Culverts – Causeways • Type of Bridge. Slab Bridge, T-beam and slab Bridge, Suspension Bridge. DAMS; • Purpose of dams • Type of Dams. Earthen Dam & Gravity Dam.	20
05	QUANTITY MEASUREMENT Procedure Of Estimating Quantities; • Introduction – Main items of work – calculation of quantities of earth work, stone masonry, brick masonry, plastering, cement concrete, R.C.C., Doors, Windows, Flooring, White Washing, colour washing, Distempering and their units.	15

	Rate Analysis; • Factors affecting rates – importance – Materials for different items of work – Rates of materials and labour – analysis of Rates for cement concrete, R.C.C., brick masonry, Stone masonry, Hollow block masonry, Plastering, Painting, Flooring, Road works, Sanitary Works, Water supply works and Electrical works.	

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DIPLOMA IN MINE SURVEYING

SEMESTER: FOURTH SCHEME: July2012

COURSE CODE: 402 PAPER CODE: 6616

NAME OF COURSE: **ELEMENTS OF CIVIL ENGINEERING**

COMMON WITH PROGRAM (S):

REFERENCE BOOKS

1. Arunachalam N., 'Basic Civil Engineering', Pratheeba Publishers, Coimbatore, 2000

2. Ramesh Babu V., 'Basic Civil Engineering', Anuradha Agencies, Kumbakonam, 2001
3. B.N.Dutta 'Estimating and Costing' S.Dutta and Company, Luuknow

RAJIV GANDHI PROUDYOGIKI VISHWAVIDYALAYA, BHOPAL

DIPLOMA IN MINE SURVEYING

SEMESTER: FOURTH SCHEME: July 2012

COURSE CODE: 403 PAPER CODE:6617

NAME OF COURSE: **MINE SURVEYING - II**

COMMON WITH PROGRAM (S):

RATIONALE

Students have to grownup his knowledge every aspect of mine surveying. He is familiar with modern survey equipment, that's used in mine. Different type of metal mine surveying helps the students to seek the job in metal mines. Knowledge of total station help the students to do some type of consultancy work of survey to stand in today's needs.

On this course student will be able to:

7. Handle the modern survey equipment Like Total station, EDM, Gyrotheodolite.
8. Conduct correlation survey by various methods.
9. Conduct Stope surveying.
10. Develop skills of recording and plotting surveyed data.
11. Solve problems related to dip, drift, borehole and faults.
12. Find the azimuth by different method.
13. Understood the photography surveying.

RAJIV GANDHI PROUDYOGIKI VISHWAVIDYALAYA, BHOPAL

DIPLOMA IN MINE SURVEYING

SEMESTER: FOURTH SCHEME: July 2012

COURSE CODE: 403 PAPER CODE:6617

NAME OF COURSE: **MINE SURVEYING - II**

COMMON WITH PROGRAM (S):

SCHEME OF STUDIES

S. No.	TOPICS	THEORY (HOURS)
12.	RECTANGULAR COORDINATE SYSTEM	15
13.	STOPE SURVEYING	10
14.	CORRELATION SURVEY	10
15.	DIP & STRIKE, DRIFT, BOREHOLE AND FAULT PROBLEM	15
16.	INTRODUCTION TO AERIAL PHOTOGRAPHY	10
17.	FUNDAMENTALS OF FIELD ASTRONOMY	10
18.	INTRODUCTION TO MODERN SURVEY TECHNIQUES	20
	TOTAL	90

RAJIV GANDHI PROUDYOGIKI VISHWAVIDYALAYA, BHOPAL

DIPLOMA IN MINE SURVEYING

SEMESTER SEMESTER: FOURTH SCHEME: July 2012

COURSE CODE: 403 PAPER CODE:6617

NAME OF COURSE: **MINE SURVEYING - II**

COMMON WITH PROGRAM (S):

Lectures: **6** Hrs. per week

CONTENT DETAILS

S. No.	Course Contents	Hrs of Study
01.	**RECTANGULAR COORDINATE SYSTEM** 1.1 Definitions; Latitudes & Departures. 1.2 Partial Latitude and Partial departures. 1.3 Calculation of Partial Latitude and Partial Departures 1.4 Total Latitude and Total Departures 1.5 Calculation of Total Latitude and Total Departures 1.6 Calculation of length & bearing from total coordinates. 1.7 Closing error and distribution , Gale's traverse table 1.8 Calculation of Area by Partial coordinate 1.9 Calculation of Area by total coordinates Methods 1.10 National grid system. 1.11 To join colliery survey with National Grid Line	15
02.	**STOPE SURVEYING**	10

	2.1 Definition and Introduction, purpose of stope survey. 2.2 Methods of stope surveying for flat, moderate inclined of steeply inclined ore deposits.	
03.	**CORRELATION SURVEY** 3.1 Purpose of correlation survey. 3.2 Classifications of methods of orientation. 3.3 Direct methods of traversing. 3.4 Assumed bearing method (Two shaft method). 3.5 Exact alignment method. 3.6 Approximate alignment method. 3.7 Wiessqudrilateral method. 3.8 Precise magnetic method. 3.9 Gyrotheodolite method. 3.10 Correlation with national grid and local scale factor.	10

RAJIV GANDHI PROUDYOGIKI VISHWAVIDYALAYA, BHOPAL

DIPLOMA IN MINE SURVEYING

SEMESTER: FOURTH SCHEME:
July 2013
COURSE CODE: 403 PAPER CODE:6617
NAME OF COURSE: **MINE SURVEYING - II**

COMMON WITH PROGRAM (S):

Lectures: **6** Hrs. per week

CONTENT DETAILS

04.	**DIP & STRIKE, DRIFT, BOREHOLE AND FAULT PROBLEM** 4.1 Definition, fault, normal, reverse and transcurrent fault, fault plane, hade of fault, throw, want, heave, excess. 4.2 Numerical problems on drift, fault, and borehole.	15
05.	**INTRODUCTION TO AERIAL PHOTOGRAPHY** 5.1 General Principle. 5.2 Field of application, Vertical and oblique photographs. 5.3 Preparation of photographical maps by simple methods.	10
06	**FUNDAMENTALS OF FIELD ASTRONOMY** 6.1 Spherical trigonometry, Latitude and longitude of a place. 6.2 Definitions of astronomical terms , 6.3 Coordinate systems to define the position of a heavenly body. 6.4 Determination of the azimuth of a line by extra meridian observation on the sun.	10

07	**INTRODUCTION TO MODERN SURVEY TECHNIQUES** 7.1 Digital theodolite , electronic distance measuring equipment , 7.2 Total station , 7.3 Gyrotheodolite 7.4 GIS & GPS 7.5 Remote Sensing	20

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DIPLOMA IN MINE SURVEYING

SEMESTER: FOURTH SCHEME:
July 2012
COURSE CODE: 403 PAPER CODE:6617
NAME OF COURSE: **MINE SURVEYING - II**

COMMON WITH PROGRAM (S):

Practical: **4** Hrs. per week

LIST OF EXPERIMENTS

	Name of Experiment	HRS OF PRACTICAL
	1. Calculation of length & bearing from total coordinates. 2. Closing error and distribution , Gale's traverse table 3. Explain the co- planning of co-relation. 4. Explain the weissbach triangle method of co-relation. 5. Explain wiessqudrilateral method of co-relation. 6. Determination of the azimuth of a line by extra meridian observation on the sun. 7. Demonstration of electronic distance measuring equipment 8. Demonstration of Digital theodolite 9. Demonstration of Total station 10. Demonstration of Gyrotheodolite.	**30**

RAJIV GANDHI PROUDYOGIKI VISHWAVIDYALAYA, BHOPAL

DIPLOMA IN MINE SURVEYING

SEMESTER: FOURTH SCHEME: July 2012

COURSE CODE: 403 PAPER CODE:6617

NAME OF COURSE: **MINE SURVEYING - II**

COMMON WITH PROGRAM (S):

REFERENCE BOOKS

1. **Surveying & Levelling Part I & II** **T.P. Kanetkar & S.V.Kulkarni**
2. **Surveying & Levelling Part I & II** **Dr.B.C.Punamiya**
3. **Mine Surveying Vol. - I II & II** **S. Ghatak**
4. **Surveying & Levelling** **N.N.Basak**
5. **Text Book of Surveying** **S.K.Husain**
6. **Higher Surveying** **A.M.Chandra**

RAJIV GANDHI PROUDYOGIKI VISHWAVIDYALAYA, BHOPAL

DIPLOMA IN MINE SURVEYING

SEMESTER: FOURTH SCHEME:
July 2012
COURSE CODE: 404 PAPER CODE: 6618
NAME OF COURSE: **MINE MANAGEMENT & ECONOMICS**

COMMON WITH PROGRAM (S):

RATIONALE

A Good Surveyor should be a good economist with management. The back bone of any Industry is economics and such Industrial Economics is included as a subject in many branches of Engineering.

In Mining Industry Economics is not applied only to higher level i.e. planning of faces in the district for production. It is necessary for a diploma pass out in mine surveying to have the Elementary knowledge of Management & Economics, its general terminology and definitions and there applications to mining operations.

OBJECTIVES:

After undergoing the course of study the student shall be able to

1. Knowledge of good management.
2. Evaluation of work with economics.
3. Knowledge of booking.
4. Calculate the different type of reserve.

RAJIV GANDHI PROUDYOGIKI VISHWAVIDYALAYA, BHOPAL

DIPLOMA IN MINE SURVEYING

SEMESTER: FOURTH SCHEME:
July 2012
COURSE CODE: 404 PAPER CODE: 6618
NAME OF COURSE: **MINE MANAGEMENT & ECONOMICS**

COMMON WITH PROGRAM (S):

SCHEME OF STUDIES

S. No.	TOPICS	THEORY (HRS.)
19.	*MANAGEMENT*	15
20.	GENERAL ECONOMICS	15
21.	MINE ECONOMICS	15
22.	*FUNDAMENTAL OF BOOK KEEPING*	10
23.	*SAMPLING, SALTING, RESERVE ESTIMATION*	20
24.	*VALUATION & DEPRECIATION*	15
	TOTAL	90

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DIPLOMA IN MINE SURVEYING

SEMESTER: FOURTH SCHEME:
July 2012
COURSE CODE: 404 PAPER CODE: 6618
NAME OF COURSE: **MINE MANAGEMENT & ECONOMICS**

COMMON WITH PROGRAM (S):

Lectures: **6** Hrs. per week

CONTENT DETAILS

S. No.	Course Contents	Hrs of Study
01	*MANAGEMENT* • *Principles of scientific management.* • Types of business organizations, organization of Coal India Ltd. • Supervision qualities of good supervisors, • Leadership, functions of industrial leadership, delegation of responsibility • Principles of time study, • Trade unions, their functions. • Strikes and lockouts	15
02	GENERAL ECONOMICS: • Economic Terms - Wealth, Value, Goods, Price • Wants-wants and economic activities, classification of wants. a. Law of diminishing utility. b. Law of equi-marginal utility • Utility- Meaning, measurement, marginal and total Utility. • Demand- Definition, demand Schedule and demand curve. Law of demand curve. i) Extension and contraction in demand. ii) Increase and decrease in demand. iii) Elasticity of demand. • Supply- Supply price, Supply Schedule, Supply curve, Supply function, Law of supply, Elasticity of supply. • Capital- Definition, Characteristics of capital. Importance and function of capital • Money- Definition of money, Function of money, Classification of money	**15**
03.	MINE ECONOMICS: • Mineral Industry- Role in national economy. a. Indian Mineral resources and their statistics. b. Mineral Policies. c. Conservation of minerals including Coal Company.	**15**

	• Constitution of companies under companies Act. a. Types of companies. b. Private and public sectors, merits and demerits. (a) Govt. Undertakings. • Labour - Efficiency of labour. a. Labour welfare. b. Social securities. c. Trade Unions.	

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DIPLOMA IN MINE SURVEYING

SEMESTER: FOURTH SCHEME:
July 2012
COURSE CODE: 404 PAPER CODE: 6618
NAME OF COURSE: **MINE MANAGEMENT & ECONOMICS**

COMMON WITH PROGRAM (S):

Lectures: **6** Hrs. per week

CONTENT DETAILS

04	FUNDAMENTAL OF BOOK KEEPING: • Basic principles of Book Keeping. The journal. Double entry system. • The books of account. The balance of various ledger pages. Trail balance. • Trading, Working or Manufacturing Account. Suspence Account. Profit & Loss Account. Balance Sheet.	**10**
05	*SAMPLING* • Principles of Sampling, Methods, Error, Selection of Sampling procedure for Particular deposits. • Preparation of sample, Coning and Quarter Splitting methods of reduction of samples. *SALTING* • Salting method of salting, salting methods by mistakes/errors. Purposeful salting procedures, Safe procedures for collecting sample to guard against purposeful salting. • Average stoping width, willing width of mining, clear width, Average width, Simple average, Weightage average, Mean values, Mining grade of ore, economical grade, cut off grade of ores, cut off grades of samples and stops. RESERVE ESTIMATION: • Type of reserve, different method of reserve estimation.	**20**
06	*VALUATION & DEPRECIATION:* • Valuation methods of valuation by different methods of annuity. Calculation of different annuities. • Methods of depreciation and calculation of Redemption values. • Main valuation methods of mining. Property which under production. Valuation under different methods. • Report Of valuation of small mining property	**15**

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DIPLOMA IN MINE SURVEYING

SEMESTER: FOURTH SCHEME:
July 2012
COURSE CODE: 404 PAPER CODE: 6618
NAME OF COURSE: **MINE MANAGEMENT & ECONOMICS**

COMMON WITH PROGRAM (S):

REFERENCE BOOKS

1. An Introduction to Mineral Economics- K.K.Chatterjee,

New Age International pub, 1993

2. Mineral Economics - R. K. Sinha and N. L. Sharma,

Oxford & IBH Pub. Co.

3. Indian Bureau of Mines, Indian Minerals Year Book, The University of Michigan, 2002

4 Industrial Organisation and Engineering Economics-. T.R.Banga and S.C.Sharma,

Khanna Pub

RAJIV GANDHI PROUDYOGIKI VISHWAVIDYALAYA, BHOPAL

DIPLOMA IN MINE SURVEYING

SEMESTER: THIRD
July 2012

SCHEME:

COURSE CODE: 405

PAPER CODE:

NAME OF COURSE: **INDUSTRIAL TRAINING**

COMMON WITH PROGRAM (S):

Practical: **2** Hrs. per week

RATIONALE

On this duration student will go to mines and visit the surrounded area of the surface & belowground. This visit helps the student to properly understand the curriculum content of fourth semester. Now the student will make a plan with neat sketch & scale by the use of CAD software. They also use the mining symbol of mining terminology of the coal mines & metal mines.

After undergoing the course of study the student shall be able to

1. Make a field note of mining working process.

5. Assign a problem properly with the practical experiences.
6. Know the storages & maintenance of plan.
7. Affiliate the problem with depth.

RAJIV GANDHI PROUDYOGIKI VISHWAVIDYALAYA, BHOPAL

DIPLOMA IN MINE SURVEYING

SEMESTER: THIRD
July 2013

SCHEME:

COURSE CODE: 406

PAPER CODE:

NAME OF COURSE: **Project**

COMMON WITH PROGRAM (S):

Practical: **6** Hrs. per week

RATIONALE

A Project for Mine Surveyor student is necessary to execute any work with proper mode. Project is the scientific way of working to reach the greater depth of any topic. On this duration student is completing final session of his branch, where screening & ascertain knowledge have to require. Student will go to mines & visit the mining method of digging, winning, transporting or hauling, need or application for mining safety.

After undergoing the course of study the student shall be able to

1. Make the mine plans on plan sheet.

8. Make the mine plans with use of cad application.
9. Calculate the area by use of levelling & contouring method.
10. Calibrate report of surveying Instrument.

Note: - Project should be going on the supervision of Lecturer, H.O.D. & Field Officer. Student will also choose the topic of project from the current use of mining method, instrument or any appliances or method, which is use on future.